

Screening pipeline

PubChemQC, first shard
IDs 1–24,999

with a B3LYP@PM6 record
17,458 molecules

B3LYP/6-31G**/PM6
HOMO, LUMO, gap, dipole

Scharber model, AM1.5G
 J_{SC} from the real spectrum

physical gates
 $E_{CT} \leq E_g$, $V_{OC} > 0$, $\Delta E \geq 0.3$ eV

admissible pairs
25 complementary, of which 16 strict

*41 of 17,458 molecules (0.23%) absorb in
1.1–2.2 eV; 25 of those clear the physical gates,
and the 16 strict pairs are a subset of those 25*

Validity gates (this work)

ranked by $PCE_{SAScore}$
top ranks include $MgCO_3$, CaO_3S_2

chemical-validity filter
R1–R6: fragments, metals, rings,
aromaticity, size, conjugation

chemically valid
11 complementary, 5 strict

stability screen
azide / nitroso / diazo

valid, stable, score > 0
4 complementary, 1 strict

R7 diagnostic (this work)
o-quinone tautomers flagged
robust lead: molecule 9168

ranked output